

ON THE UNDECIDABILITY OF MEDVEDEV'S LOGIC

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LL(A)M-A seminar

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Institute for Logic, Language and Computation

Outline of today

The talk will be split in two sessions:

1. (45 minutes) Undecidability of Medvedev's logic, and related results + Q & A.

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1. (45 minutes) Undecidability of Medvedev's logic, and related results + Q & A.
2. (15 minutes) How these results were obtained + where to go from here.

History and context

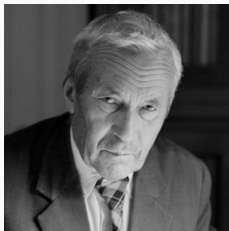


Figure 1: A. Kolmogorov, (1903–1987).

“Intuitionistic Logic is the logic of finding solutions to problems”

Kolmogorov's interpretation of intuitionistic logic

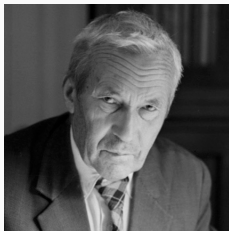


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“Intuitionistic Logic is the logic of finding solutions to problems”

This was **much before** any semantics for the logic existed.

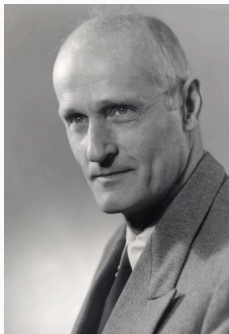


Figure 2: S. C. Kleene (1909 – 1994).

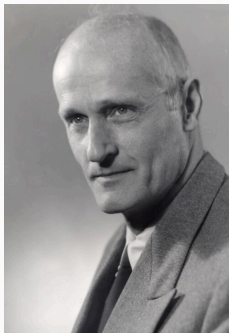


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Kleene proposed his **realizability interpretation** in 1945, along similar lines. It was shown by Rose (1953) however that **intuitionistic logic was not complete with respect to it**.

Medvedev's problems

In 1962, Yu Medvedev publishes a paper, returning to the roots of Kolmogorov's interpretation:

1. By a finite problem we mean, in the present paper, any problem the solution of which is an element of some previously known nonempty finite set F of admissible possibilities. As examples of finite problems we may use the problems of determining the truth or falsity of propositions: here the set F consists of two elements, "true" and "false", and that one of the two which is the truth value of the proposition serves as the solution. Chess problems also may be considered as finite under the appropriate interpretation.

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The goal was to understand intuitionistic logic through problems. But Medvedev quickly observed that when using the full signature the resulting validities were much more than what we now call IPC.

In modern terms, we typically formulate the logic using Medvedev frames; for I an arbitrary set:

$$M_I := (\mathcal{P}(I) \setminus \{I\}, \supseteq),$$

and

$$\text{Med} := \text{Log}(\{M_I : I \text{ a finite set } \}).$$

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1. Levin (1969) showed that **Med** has the disjunction property and maximal with that property – disproving a conjecture of J. ukasiewicz.
2. Prucnal (1976, 1979) showed that Medvedev's logic is structurally complete, answering a question as to the existence of logics with disjunction property and structurally complete, of H. Friedman.

This made it a very attractive logic to study from a constructive point of view. But at this point in time, **no axiomatization of it was known.**



Figure 4: L. Maksimova (1943 - 2025); D. Skvortsov (1952 - 2025); V. Shehtman (1953 -)

Theorem (MDS, 1979)

The logic Med is not finitely axiomatizable.

Skvortsov and Shehtman obtained several results, and introduced several new questions:

1. The logic Skv of [infinite problems](#), characterized by all Medvedev frames. Shown by Skvortsov (1972) to be recursively axiomatizable. Further study was made together with Shehtman (1982). Its equality with Med remained open.

Chasing Medvedev's logic

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1. The logic Skv of [infinite problems](#), characterized by all Medvedev frames. Shown by Skvortsov (1972) to be recursively axiomatizable. Further study was made together with Shehtman (1982). Its equality with Med remained open.
2. A modal counterpart of Med , the logic $Med_{\Box}$ was introduced by Shehtman (1990), and proven not finitely axiomatizable, its decidability also remained open.

At the same time, Medvedev's logic kept appearing in other, seemingly unexpected, places...

Ubiquity of Medvedev logic

The logic, and some close relatives, kept appearing:

1. **(Topology and Geometry)**: The logic **Cheq** of chequered subsets (van Benthem et.al. 2003), bears a close relationship with **Med**, and was introduced to analyze Euclidean space using modal logic.

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Many, many more contributions: generalized Medvedev logics (Xiao, 2024), approaches via tense modalities (Holliday, 2017), as well as studies into the related logics (Fontaine, 2006; Shatrov, 2008; Litak, 2018).

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(Except for Søren for a while. But recently he too was convinced.)

Anatomy of the logic

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It is well-known that the fragment $\text{IPC}_{\wedge, \rightarrow, \perp, \top}$ is **locally tabular**.

Axiomatizing with canonical formulas

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The idea of such formulas is algebraically very transparent:

Proposition

For each Heyting algebra H , $H \models \alpha(P, D)$ if and only if there is an injective map $\iota: \mathbf{Up}(P) \hookrightarrow H$, which is a $(\wedge, \rightarrow, \perp, \top)$ -homomorphism, and for each pair $(U, V) \in D$, preserves the join.

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Upshot: One only needs to consider canonical formulas.

Partial maps

This can be turned into a more geometric criterion:

Definition (Cofinal partial p-morphisms)

A **partial** map $f: P \rightarrow Q$ between posets is called a **cofinal partial Esakia morphism** if the following conditions hold:

- (Order) If $p, p' \in \text{dom}(f)$ and $p \leq p'$ then $f(p) \leq f(p')$.

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Partiality turns out to be one of the crucial pieces. They will make **constructing a map easier**.

Definition (Closed domain condition)

Let Q be a finite rooted poset, P a poset, and $\mathfrak{D}$ a (possibly empty) collection of antichains in Q . A cofinal partial Esakia morphism $f: P \rightarrow Q$ is said to satisfy the **closed domain condition** (CDC) with respect to $\mathfrak{D}$ if:

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Closed domains provide us some additional strength to offset the weakness of partial maps: contrapositively, they force the existence of points in the domain.

Reducing to existence of a map

Definition (Medvedev cover)

Given P a finite rooted poset, and $D \subseteq (P)^2$ we denote by $\text{Cov}_{\text{Med}}(P, D)$ the statement: there is some $n \in \omega$ and an onto cofinal partial Esakia morphism $M_n \rightarrow P$ satisfying the CDC for $\mathfrak{D}_D$.

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Theorem (Validity of canonical formula for Medvedev frames)

For each canonical formula $\alpha(P, D)$, we have:

$$\alpha(P, D) \notin \text{Med} \iff \text{Cov}_{\text{Med}}(P, D).$$

How do we prove undecidability?

Tilings

The Tiling Problem

- A **Wang tile** is a square with colored sides.
- **The tiling problem:** Given a finite set of tiles $\mathcal{W}$, does there exist a tiling $\tau : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathcal{W}$.

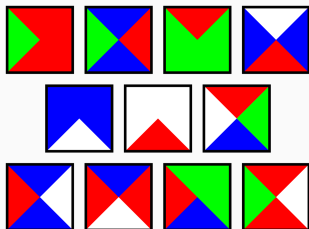


Figure 5: Wang tiles

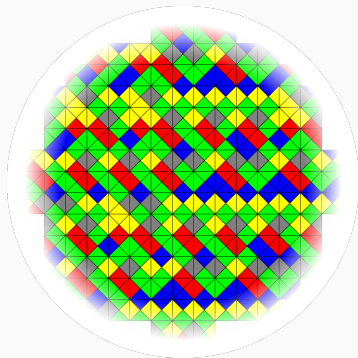


Figure 6: A tiling $\tau : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathcal{W}$

Figures taken from: https://en.wikipedia.org/wiki/Wang_tile

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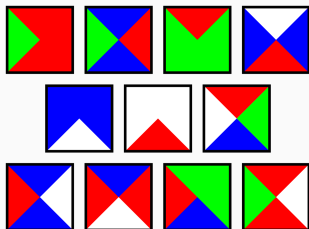


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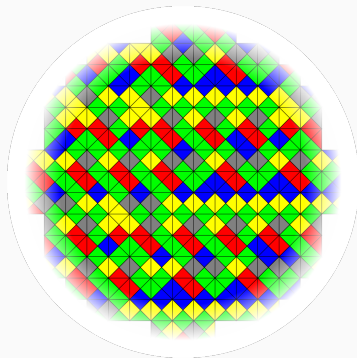


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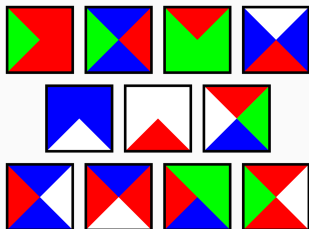


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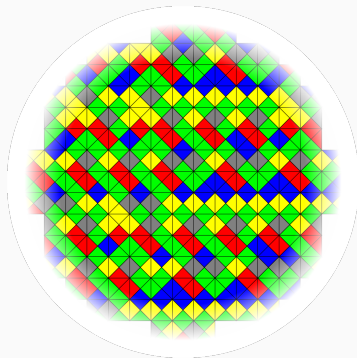


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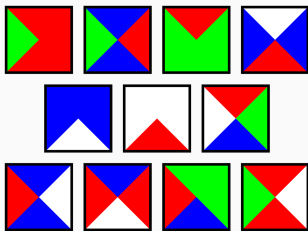


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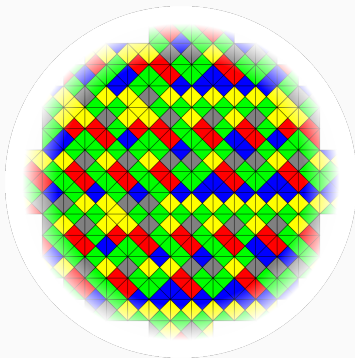


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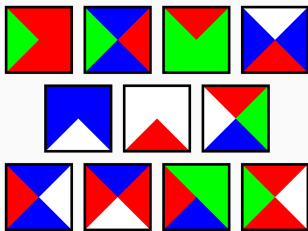


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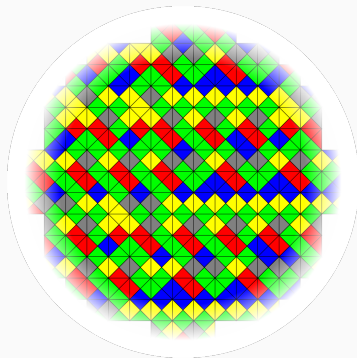


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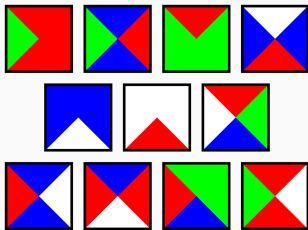


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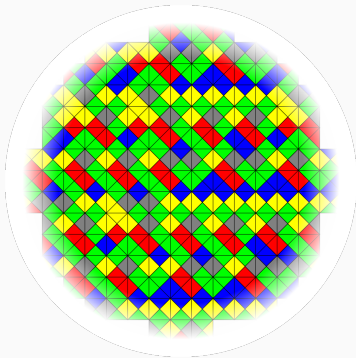


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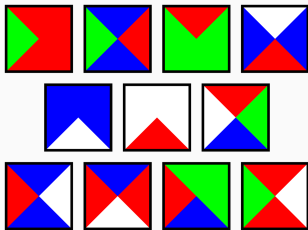


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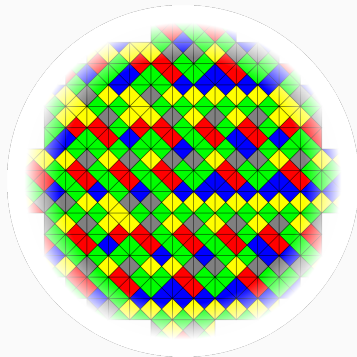
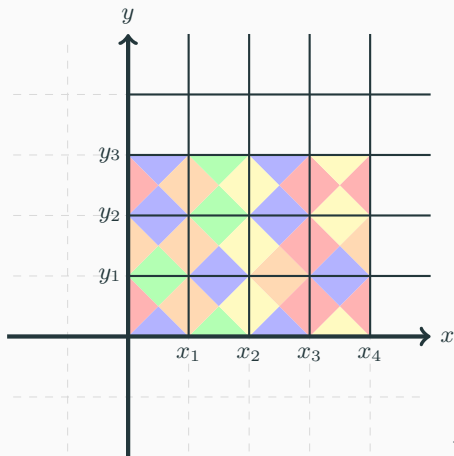


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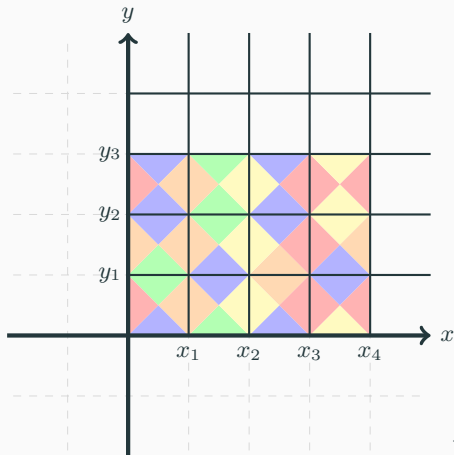
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Recall: $\alpha(P, D) \notin \text{Med} \iff \text{Cov}_{\text{Med}}(P, D).$

Strategy: Given $\mathcal{W}$, construct poset $P_{\mathcal{W}}$ and $D_{\mathcal{W}} \subseteq \text{Up}(P_{\mathcal{W}}) \times \text{Up}(P_{\mathcal{W}})$ with

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$$\boxed{\exists \tau : \mathbb{Z}_m \times \mathbb{Z}_n \rightarrow \mathcal{W} \iff \text{Cov}_{\text{Med}}(P_{\mathcal{W}}, D_{\mathcal{W}}).}$$

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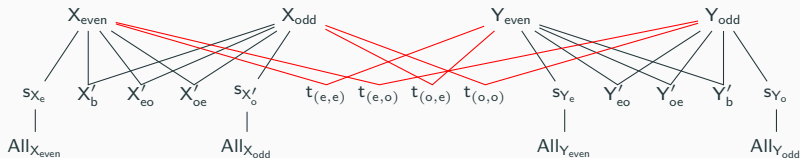
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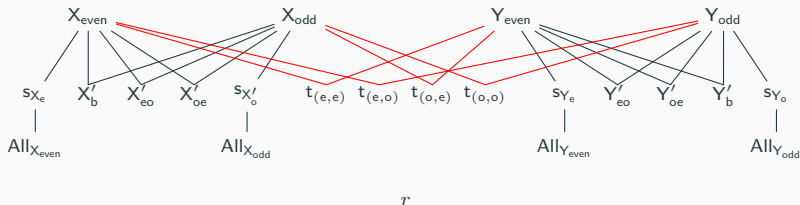


CDCs (D_W) ensure:

- Doubletons in $\text{dom} f$
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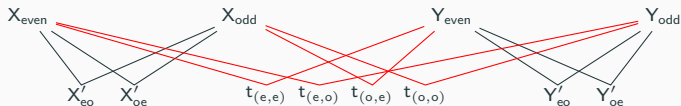


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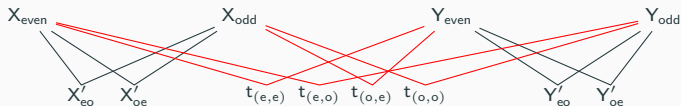
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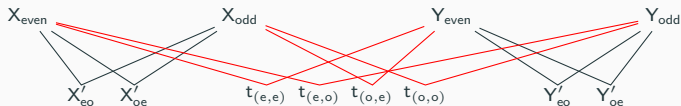
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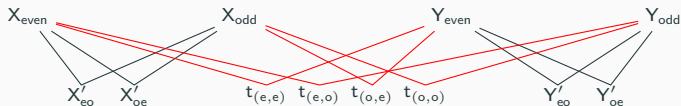
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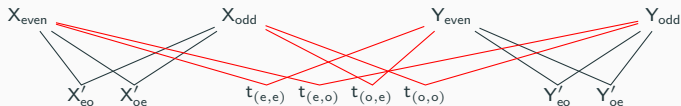
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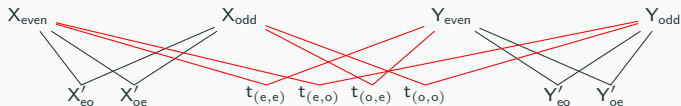
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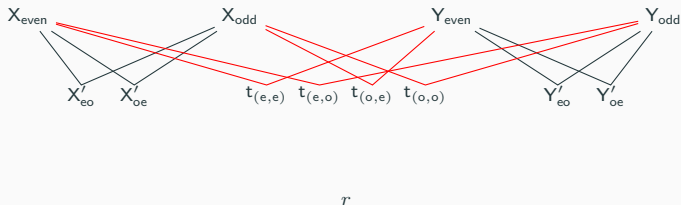
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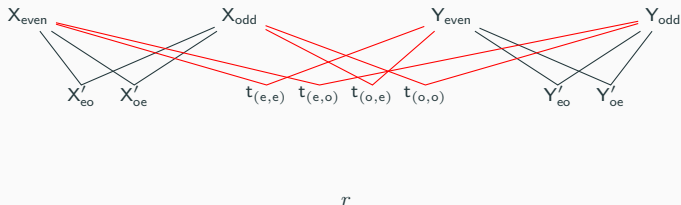
$$\begin{aligned} \min f(\uparrow x) \cap (U \setminus V) &\neq \emptyset, \\ \min f(\uparrow x) \cap (V \setminus U) &\neq \emptyset \end{aligned} \Rightarrow \min f(\uparrow x) \cap (U^c \cap V^c) \neq \emptyset$$

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Take-aways

- Partiality.
- CDCs can **force** $\{a, b\} \in \text{dom } f$;
· and **impose restrictions** on $\{a, b, c\} \notin \text{dom } f$.
- Periodicity through finiteness.
- Meets and distributivity.

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Main theorems

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Med is undecidable.

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Skv is undecidable.

Proof.

The tiling proof covers Med and Skv uniformly: for Skv we encode the tiling problem, and for Med the periodic one. $\square$

Corollary

For every aperiodic tile set $\mathcal{W}$:

$$\alpha(P_{\mathcal{W}}, D_{\mathcal{W}}) \in \text{Med} \setminus \text{Skv}$$

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But *why* are Med and Skv undecidable?

Directions, repercussions, perspectives

Intuitionistic and modal logic:

- Theorem: **Modal Med** and **modal Skv** are undecidable.
- *Direction:* **Interval** of undecidable logics?
- *Direction:* Robust decidability, revisited?

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Wir müssen verstehen. Wir werden verstehen.

Yankov axiomatizability

Shehtman (1990) noted that the question of **axiomatizability by Yankov formulas** was a potential avenue for studying Med. The following are still open, and quite interesting problems:

Let Y_{Med} be the logic

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*Is $Y_{\text{Med}} = \text{Med}$? Is Y_{Med} recursively axiomatizable? Does it have the FMP? Is the problem of recognition of finite **Med** algebras decidable^a?*

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In **geometric terms**, this amounts to deciding whether the existence of specific **open colorings** of a simplex (where a poset encodes the allowed colors) is a decidable problem.

Degrees of FMP

Such questions could be interesting from other points of view:

Definition (Degree of FMP)

A superintuitionistic logic L is said to have **degree of the finite model property** κ for κ a cardinal if and only if the following set has cardinality κ^a :

$$\deg_{\text{FMP}}(L) := |\{L_0 \in \text{Ext}(\text{IPC}) : L_0 \text{ has the same finite frames as } L\}|.$$

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Problem

What is the degree of FMP of Med?

If $Y_{\text{Med}} = \text{Med}$, the degree would be 1. Potentially it could be very large.

Having obtained these results, it might be interesting to reconsider some older logics.

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The logic Realiz is **propositional realizability logic** – inspired by the work of Kleene, and subsequently studied by several authors. It has close historical ties, and shares some non-trivial axioms with Med. Nevertheless:

Problem

Is Realiz decidable? Does it have the FMP? Is it recursively axiomatizable?

Thank you!
Questions?

Methodology

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4. Why did we make this available?
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